# Manufacturing Downtime Root Cause Analysis Report

## Executive Summary

The R101 Wing Positioning & Fixturing Robot is currently in a **CRITICAL** state, as indicated by sensor data and anomaly detection. Key sensor readings suggest potential mechanical or operational issues, and recent events highlight a servo overload incident. Immediate attention is required to assess and address the machine's health.

## Machine Overview

- **Machine ID**: R101
- **Equipment**: Wing Positioning & Fixturing Robot
- **Plant / Location**: Not specified in the evidence
- **Criticality**: High (based on current state and operational role in wing assembly)
- **Analysis Time**: Timestamped data up to 2026-08-08

## Current Machine Condition

The machine is in a **CRITICAL** state, as indicated by the following sensor observations:

- **Joint Temperature**: 42.64°C (elevated but not critical)
- **Motor Current**: 6.03 A (elevated, potentially indicating increased load or resistance)
- **Joint Vibration**: 0.131 mm/s (within acceptable range)
- **Encoder Position**: 89.96° (appears normal)
- **Torque**: 20.63 Nm (elevated, suggesting mechanical resistance or load issues)

An anomaly detection system has flagged sensor data as abnormal, further supporting the need for investigation.

## Historical Analysis

- **Event FE0012**: A servo overload due to excess load on the positioning axis occurred on 2026-06-29. This may have contributed to the current state if not fully resolved.
- **Recent Events**: The machine successfully completed production cycles and transitioned between batches on 2026-08-07, with no immediate operational failures reported.
- **Maintenance**: No evidence of recent maintenance activities or corrective actions was provided, despite the machine's critical state.

## Root Cause Analysis

- **Primary Root Cause**: Likely **mechanical resistance or load imbalance** on the positioning axis.
- **Supporting Evidence**:
- Elevated motor current (6.03 A) and torque (20.63 Nm) suggest increased mechanical resistance.

• Historical servo overload event (FE0012) indicates prior load-related issues.

• Anomaly detection flagged sensor data as abnormal.

• **Contributing Factors**:

• Lack of recent maintenance or corrective actions.

• Potential wear or degradation of components such as bearings or gearboxes, as these are common failure modes for this machine type.

## Operational Impact

The machine's critical state poses a significant risk to production continuity. If unresolved, the following impacts are likely:

• Reduced reliability and potential for unplanned downtime.

• Delays in wing assembly processes, affecting overall production schedules.

• Increased maintenance costs if the issue escalates to a major failure.

## Recommendations

### *Immediate Actions*

1. Perform a detailed inspection of the positioning axis, focusing on bearings, gearboxes, and mechanical alignment.
2. Verify sensor health and recalibrate encoders to ensure accurate readings.
3. Reduce operational loads temporarily to prevent further strain on the machine.

### *Short-Term Actions*

1. Conduct a root cause investigation into the servo overload event (FE0012) to determine if it was fully resolved.
2. Implement vibration and motor current trend monitoring to detect early signs of mechanical degradation.
3. Schedule preventive maintenance to address potential wear and tear.

### *Long-Term Improvements*

1. Establish a routine maintenance program with detailed logging of activities, as recommended in the operator manual.
2. Upgrade anomaly detection systems to provide more actionable insights and early warnings.
3. Train operators on best practices, such as avoiding abrupt acceleration with maximum payloads.

## Risk Assessment

• **Likelihood of Recurrence**: **High**

Justification: The machine is in a critical state, and no recent maintenance or corrective actions have been documented. Historical issues (e.g., servo overload) suggest recurring load-related problems.

## Confidence Level

- **Confidence Level**: **Medium**

Justification: While sensor data and historical events provide strong indicators, the lack of detailed maintenance records limits the ability to confirm the root cause with high confidence.

## Conclusion

The R101 Wing Positioning & Fixturing Robot is in a critical state, likely due to mechanical resistance or load imbalance on the positioning axis. Immediate inspection and maintenance are required to prevent further operational disruptions. Establishing a robust maintenance program and monitoring system will mitigate future risks.